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Research Article

Self-densified super-strong wood

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ABSTRACT

Lightweight structural materials with high strength and toughness are highly desirable for many advanced applications. Wood, as a sustainable structural material, is widely used in engineering due to its abundance and excellent mechanical properties. In this paper, we report a self-densification strategy to develop super-strong wood by reassembling highly aligned wood fibers as functional units and self-densified without the need for hot pressing. The resulting self-densified wood exhibits ultra-high tensile strength (496.1 MPa), flexural strength (392.7 MPa) and impact toughness (75.2 kJ/m^2), surpassing those of compressed densified wood and traditional metal materials like aluminum alloys. Notably, the self-densified wood exhibits uniform shrinkage in the cross-section while maintaining its longitudinal dimension. This characteristic leads to an order-of-magnitude enhancement in the overall mechanical performance of the wood, presenting a significant advantage over compressed densified wood. Such super-strong yet lightweight wood has great potential for application as a sustainable engineering material, replacing traditional structural materials such as metals and alloys.

1. Introduction

Recently, a novel paradigm in material design has emerged, based on ordered structures with functional units such as compositions, phases and twin crystal. By employing artificial units with specific functional properties as building blocks and arranging them in a certain spatial order in a deliberate spatial order, new materials with extraordinary characteristics can be expected (Chen and Li, 2019). In fact, the realization of special functions through the construction of ordered structures has long existed in nature, as exemplified by wood. This material design concept inspires us to use multi-scale wood fibers as functional units to develop new microstructural materials with groundbreaking mechanical properties.

Natural wood is a traditional building material in human history that has been used for millennia as a structural material. Even today, wood remains widely employed in construction, furniture, and various others fields (Chen et al., 2020a; Ding et al., 2023; Schubert et al., 2023). The mechanical properties of wood are determined by its unique hierarchical structure with pronounced anisotropy and lignocellulosic compositions, which are formed over millions of years of evolution. The rigid wood cell walls consist of highly aligned cellulose fibrils embedded in the lignin and hemicellulose matrix, giving trees essential properties needed for structural support, resisting strong wind, and providing biological functions (Ling et al., 2018; Maaß et al., 2020; Jakob et al., 2022). However, the microstructure of natural wood is not optimal from a structural material perspective. To balance mechanical strength with biological functions, certain mechanical properties are inevitably sacrificed. The strength of natural wood is significantly lower

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than that of steel (Kim et al., 2015; Frey et al., 2019) and is often inadequate for advanced engineering structures and applications when compared to synthetic materials such as metals, alloys (Dursun and Soutis, 2014; Huang et al., 2019), and engineering plastics (Chen et al., 2021).

Wood-derived cellulose nanofibers are considered as the most promising candidate for replacing some of these nonrenewable and petroleum-based synthetic fibers (such as carbon fibers and glass fibers), especially given their advantage of being environmentally friendly, renewable, and lightweight. In theory, the mechanical strength of elementary cellulose fibrils could reach 7.5 GPa (Moon et al., 2011; Yang et al., 2021), leaving significant room for further improvement of bulk wood. Song et al. (2018) reported a densification strategy involving partial delignification followed by hot pressing to develop a super-strong wood. The strength of the resulting material was improved to 560 MPa, which enabled wood to compete with steel for the first time. Such engineered wood has attracted worldwide attention and promises to be the next-generation super-strong and super-lightweight structural material.

So far, super-strong wood is fabricated through mechanical pressing perpendicular to the wood growth direction (Berglund and Burgert, 2018; Li et al., 2020b; Luan et al., 2022). Such unidirectional compression enables the wood lumina as well as the porous wood cell walls to collapse entirely, resulting in a unique microstructure: the fully collapsed wood cell walls are tightly intertwined along their cross-section and densely packed along fiber direction. The anisotropic structure further exacerbates the directional differences in the mechanical properties of wood. Compression densification significantly improves the longitudinal tensile strength of wood (compressed wood ≫ natural wood), and also enhances the transverse tensile strength when perpendicular to the compression direction (compressed wood ≫ natural wood) (Song et al., 2018; Jakob et al., 2022). However, when the load axis is parallel to the compression direction, the transverse tensile strength of compressed densified wood is considerably lower than that of natural wood, primarily due to microdefects, stress concentrations, viscoelastic behavior, and shape memory effects (Tarkow, 1968); Kyriazidou and Pesendorfer, 1999; Rautkari et al., 2010; Chen et al., 2018). Additionally, compression densified wood faces issues such as expansion and rebound along the compression direction (Marbun et al., 2023). Although hydrogen bonding or chemical treatments can partially improve dimensional stability, the issue of low transverse tensile strength in the compression direction persists, limiting its application in complex mechanical environments. Therefore, the development of a new isotropic densification technology that improves the mechanical properties and dimensional stability of wood in all directions is key to addressing this problem and will open new avenues for the broader application of wood.

In this paper, we report a self-densification strategy to transform porous natural wood directly into a dense and super-strong material without hot-pressing. Our approach involves setting free cellulose fibrils from the lignin confinement through partial delignification, then the cellulose fibrils moving inward and filling the cell lumen via lithium chloride/dimethylacetamide (LiCl/DMAc) swelling process, followed by air-drying. Unlike unidirectionally compressed densified wood, our self-densified wood exhibits uniform shrinkage in the transverse area, resulting in greatly improved mechanical properties in all directions (Figs. 1a–1c). Multi-scale aligned wood fibers, as functional units assembled with enhanced hydrogen bonds, exhibit excellent flexural performance, with the longitudinal flexural strength approaching 400 MPa. This represents a significant advantage over compressed wood, where the longitudinal flexural strength varies considerably depending on whether the load axis is aligned with the compression or non-compression direction (96 MPa vs. ~400 MPa) (Fig. 1d). The resulting sustainable, super-strong wood emerges as a strong competitor for replacing traditional structural materials such as metals and alloys.

2. Materials and methods

2.1. Chemicals and raw materials

The basswood used in this study is scientifically classified as *Tilia amurensis*. It primarily grows in Northeast China. The material was purchased from Oriental Aolong Wooden Crafts Co., Ltd. Chemicals used for removing lignin from wood were sodium hydroxide (NaOH), and sodium sulphite (Na_2SO_3), both of which were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai). The anhydrous lithium chloride (LiCl) was purchased from Xilong Scientific Co., Ltd. The solvents used included N,N-dimethylacetamide (purchased from Sinopharm Chemical Reagent Co., Ltd.) and deionized water.

2.2. Fabrication process of self-densified wood

First, natural wood blocks were immersed in a boiling lignin removal solution of mixed NaOH solution (2.5 mol/L) and Na_2SO_3 solution (0.3 mol/L) for 10 h, followed by immersed in boiling deionized water several times to remove residual chemicals. The partially delignified wood blocks were preserved in N,N-dimethylacetamide to remove residual water. Next, partially delignified wood were immersed in N,N-dimethylacetamide at 160 °C for 1 h to activate wood cellulose. Then, LiCl (the mass ratio of LiCl and N,N-dimethylacetamide is 1:9) was added and the swelling reaction was carried out at 100 °C without stirring for 1 h, followed by being placed at room temperature for 10 h. The swelling wood was washed with deionized water several times to remove residual chemicals. Finally, the swelling wood was air-dried at room temperature and shrank spontaneously to obtain the self-densified wood.

2.3. Fabrication process of compressed densified wood

First, the basswood sampleswere immersed in a boiling aqueous solution of 2.5 mol/L NaOH and 0.4 mol/L Na_2SO_3 for 10 h. Subsequently, they were repeatedly washed with deionized water to completely remove any residual chemicals. The treated wood

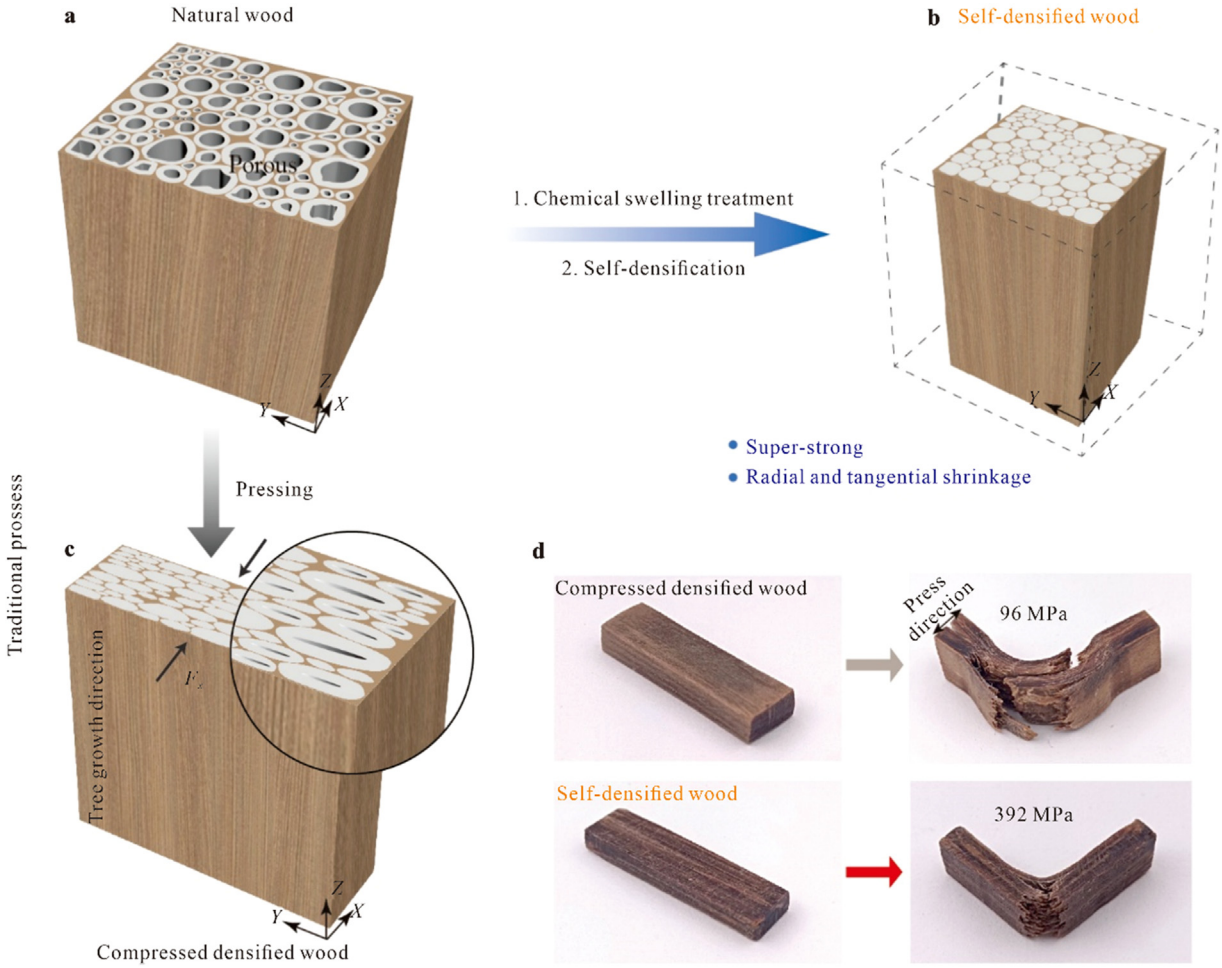

Fig. 1. Schematic of self-densification strategy to transform (a) natural wood into (b) self-densified super-strong wood without hot-pressing, compared with (c) traditional compressed densified wood. (d) Photographs of the compressed densified wood featuring one-direction reinforcement and self-densified wood featuring multi-direction reinforcement.

blocks were then subjected to compression at approximately 3 MPa in a 80 °C environment for 24 h, resulting in a highly compressed densified wood material.

2.4. Measurement and characterization

The morphologies of the partially delignified wood, the swelling wood and the self-densified strong wood were characterized by scanning electron microscope (SEM, FEI Quanta200, USA and ZEISS GeminiSEM 500, Germany). Quanta 200 SEM was used to observe micron-scale structures; GeminiSEM 500 was used to observe nanoscale structures. The wet delignified wood and swollen wood SEM samples were obtained through freeze-drying to maintain their dimensions. A platinum nanolayer was sputtered on the surface of the samples prior to SEM imaging. The X-ray diffraction (XRD) patterns of natural wood and self-densified wood were obtained by a diffractometer (Rigaku, ULTIMA-4 XRD, Japan) using Cu-Kα radiation over a range of 5°–50° at a scanning speed of 0.01°/s. The chemical changes of wood in delignification process and swelling process were studied by fourier transform infrared spectroscopy with attenuated total reflection(FTIR-ATR) (NICOLET, NEXUS870, USA) with a scanning range of 3000–500 cm^{-1}. the national renewable energy laboratory (NREL) method was used to quantitatively determine the lignin content in both natural wood and self-densified wood.

2.5. Mechanical test

The tensile, three-point bending and compressive properties of the wood samples were tested using a universal mechanical testing machine (Shimadzu, AG-X Plus 50KN, Japan) with a video non-contact axial extensometer (Shimadzu, TRViewx, Japan). Dumbbell-shaped tensile specimens were laser cut from the samples (dimension: 80 mm (length) × 2 mm (height) × 6 mm (width). The standard

samples were stretched at a speed of 5 mm/min at room temperature until they fractured. The dimensions for bending samples were 70 mm by 6 mm by 3 mm. Three-point bending tests were conducted with the span between the two bottom rollers 34 mm, and the top roller pressing down at the middle at the speed of 2 mm/min. For compression tests, the dimensions for samples were approximately 10 mm long, 10 mm wide and 8 mm thick, and the samples were compressed along the thickness direction at a speed of 1 mm/min. The charpy impact properties of the wood were tested using pendulum impact testing machine (501J-3, Shenzhen Wance Testing Machine Co., Ltd., China) and the general size of samples were 60 mm by 10 mm by 3 mm. For all the tests, at least five specimens of each group were tested, and the average values were reported with their standard deviations. The shear tests and pullout tests of wood nails were performed using a universal mechanical testing machine (Shimadzu, AG-X Plus 50KN, Japan). For shear tests, the nails were hammered into a 3-layer basswood block (thickness of each layer: 1 cm). For pullout tests, the nails were hammered into a basswood blocks with the depth of 1 cm.

3. Results and discussion

3.1. Fabrication and microstructure analysis

We synthesized the super-strong, self-densified wood via the microstructural regulation strategy. As shown in Figs. 2a and 2b, after regulation, natural wood underwent uniform shrinkage in the transverse area while maintaining its longitudinal dimension, resulting in a 79.0% reduction in volume and a nearly threefold increase in density (1.25 g/cm^3), which is similar to that of mechanically compressed wood (Fig. 1c, Fig. 2c) (Jakob et al., 2022). This transverse area shrinkage with unchanged longitudinal length was especially noticeable in irregularly shaped wood, where spheres became ellipsoids and cones sharpened (Supplementary Materials, Figs. S1 and S2).

In order to study the microstructural evolution of the self-densified wood, we used the SEM to observe the swelling and densification processes. Natural wood possesses a hierarchical cellular structure with aligned cellulose fibrils confined in the cell wall matrix (Supplementary Materials, Figs. S3–S5). The lignin and hemicellulose deposited in the cell wall are tightly bonded with cellulose fibrils through covalent and hydrogen bonds, which fix the position of the cellulose fibrils (Li et al., 2021; Chen et al., 2024). First, a common NaOH/Na_2SO_3 delignification process was used to partially remove lignin and hemicellulose from the wood cell walls (Figs. 2d and 2g, Supplementary Materials, Figs. S6–S9) (Kumar et al., 2021). The lignin content decreased from 26.6% to 12.9% due to delignification (Supplementary Materials, Fig. S10). This process freed the cellulose fibrils from the constriction of lignin, giving cellulose fibrils the ability to move more easily. Fourier-transform infrared spectroscopy (FT-IR) indicated the effective removal of lignin and hemicellulose as the characteristic peaks assigned to lignin and hemicellulose (C = O stretching vibration) significantly reduce (Supplementary Materials, Fig. S11) (Sreenivasan et al., 2011; Saravanakumar et al., 2014; Chen et al., 2020b). Then, the partially delignified wood was swollen by the LiCl/DMAc treatment. Due to the restriction of the middle lamella and residual lignin (residual lignin:12.9%, Supplementary Materials, Fig. S10), the cellulose fibrils swelled and moved evenly inward and filled the cell lumen (Fig. 2e, Supplementary Materials, Figs. S12–S15). The secondary wall and S1 layer, to some extent, guide the swelling direction, leading to significant radial expansion of plant fibers (Solhi et al., 2023). Note that the cellulose fibrils were swelling, not dissolution. No fine, disordered, or poorly aligned regenerated cellulose fibers were observed in the SEM images. The cell, nanofiber bundles, and microfibrils within the wood maintained their longitudinal alignment without significant changes. During the swelling process, the zoomed-in SEM images show that the density of cellulose microfibrils in the wood cell wall significantly decreased from 288 to 102 fibers per μm^2, demonstrating the successful swelling process (Figs. 2i and j). This process is essential to the following self-densification as it effectively exposes more hydrophilic hydroxyl groups of cellulose fibrils, which would greatly facilitate the re-formation of H-bonds between adjacent nanofibers. The crystalline structure of the self-densified wood was found to be similar to that of natural wood (cellulose I) (Wan et al., 2017), indicating that the crystalline structure of cellulose remained unchanged during the LiCl/DMAc swelling process (Supplementary Materials, Fig. S16) (Dong et al., 2022). Finally, the swelling wood was washed with distilled water and air-dried at room temperature (Fig. 2f, Supplementary Materials, Figs. S17–S21). The fibrils in wet swelling wood were covered with water molecules that bond weakly with the –OH groups of cellulose (Jin et al., 2015; Kulasinski et al., 2017; Hou et al., 2021; Zhang et al., 2021). This weak hydrogen bonding led to a looser interfacial connection between cellulose and water molecules, thereby weakened the synergistic interactions between fibers. During water evaporation, the capillary pressure in the xylem conduit caused spontaneous shrinkage of the bulk wood (Li et al., 2020a; Chen et al., 2024). As the adjacent cellulose fibrils came into contact, and stronger hydrogen bonds formed between the hydroxyl groups of cellulose molecules (CNC–CNC), ultimately resulting in a highly densified and compact structure. (Fig. 2h).

3.2. Mechanical performance

The longitudinal mechanical properties of self-densified wood not only far exceed those of natural wood, but also significantly outperform those of many widely studied compressed densified woods. Fig. 3a compares the tensile stress-strain curves of natural wood and self-densified wood. Both curves show a linear deformation behavior before tensile failure. The tensile strength of the self-densified wood is approximately 496.1 MPa, which is 9.0 times higher than that of the original wood (55.3 MPa). Fig. 3b compares the tensile strength of self-densified wood, natural wood, and compressed densified wood, a widely recognized method for enhancing strength. The results show that self-densified wood performs similarly to or even better than compressed densified wood in tensile strength. Self-densified wood exhibits exceptional specific strength, reaching 396.8 MPa/(g·cm^{-3}) (Fig. 3c, Supplementary Materials, Fig. S22) (Naskar et al., 2016; Zheng et al., 2018), making it a promising new ultra-lightweight engineering material. It not only

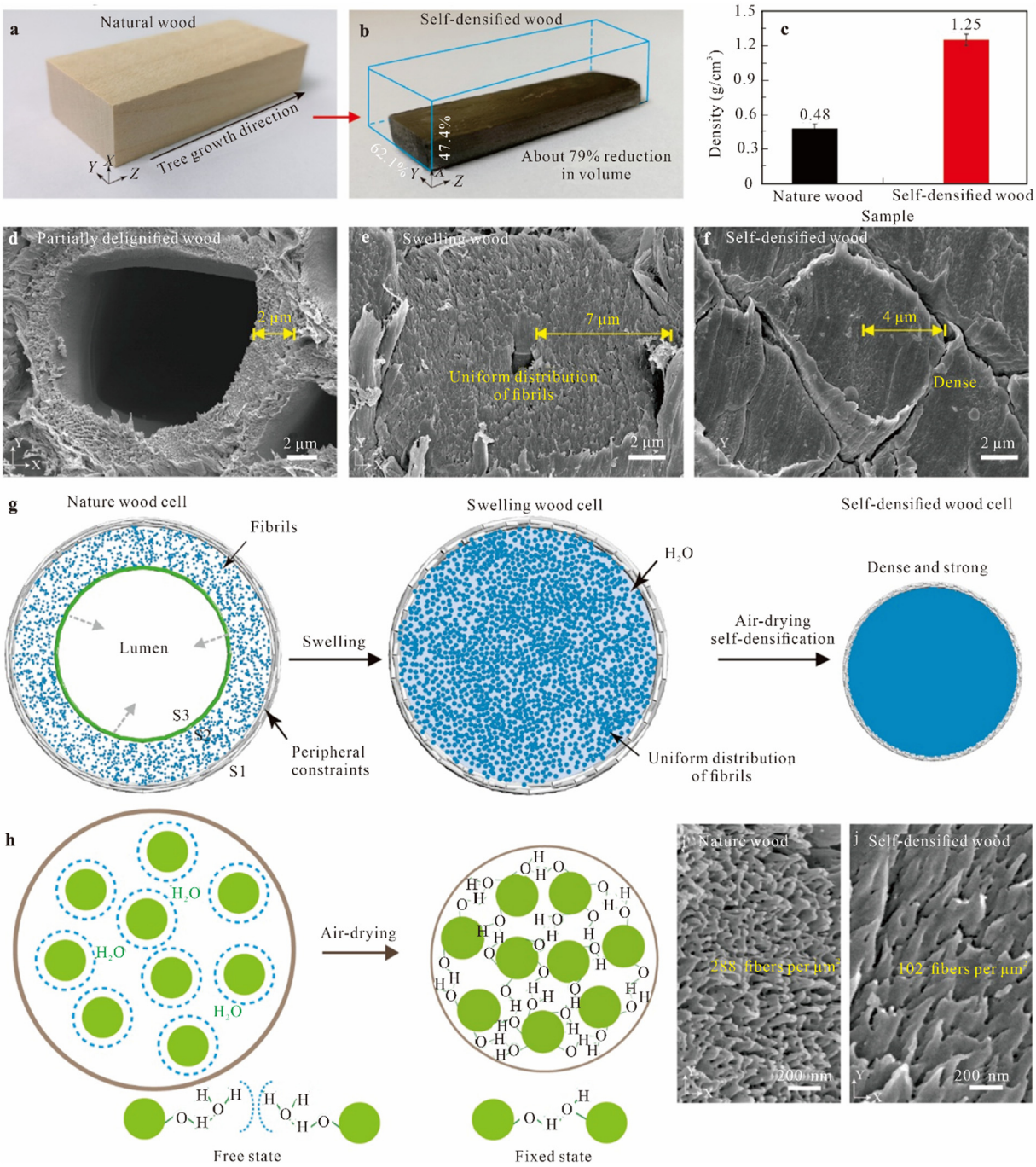

Fig. 2. Microstructure and mechanism underlying the wood self-densification process. Photographs of (a) natural wood and (b) self-densified wood, indicating significant volume shrinkage. (c) Densities of the natural and self-densified wood. Scanning electron microscopy (SEM) images of (d) partially delignified wood cell, (e) swelling wood cell, and (f) self-densified wood cell. The cellulose fibrils first moved inward and filled the cell lumen, then self-densified into a dense and solid material. (g) Schematic illustration of self-densification process (S1, S2 and S3 are the outer layer, middle layer, and inner layer of the secondary wall of wood cell walls, respectively.). (h) Schematic illustration showing hydrogen bonding formation between adjacent cellulose fibrils during air-drying. The SEM images of (i) partially delignified wood cell wall featuring fibrils density of 288 fibers per μm^2, and (j) intermediate swelling wood cell wall featuring fibrils density of 102 fibers per μm^2.

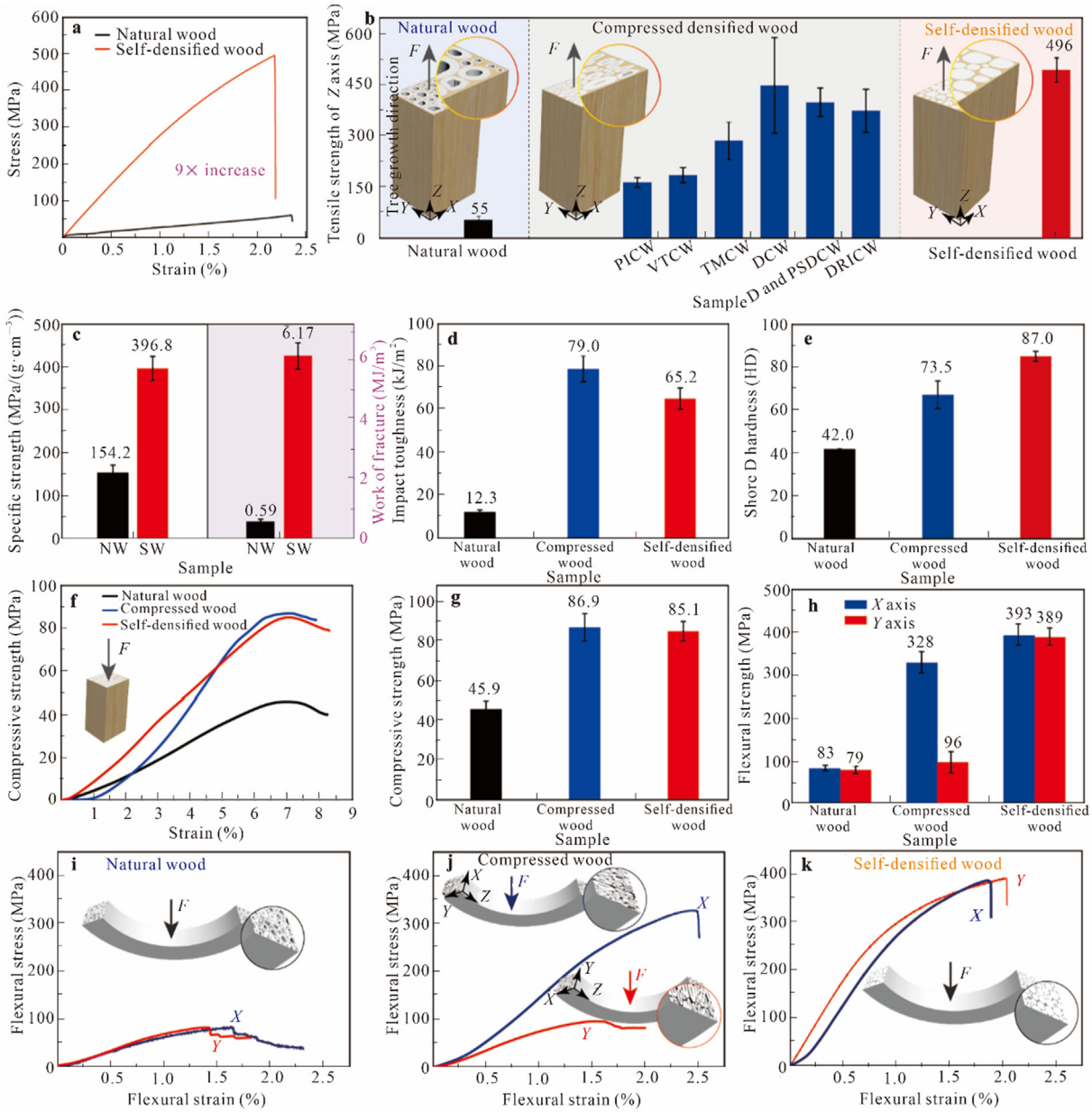

Fig. 3. Superb longitudinal mechanical properties of self-densified wood. (a) Tensile stress-strain curves of natural and self-densified wood. (b) Specific tensile strength of the resulting self-densified wood compared to natural wood and various mechanical-compressed wood. Polymer-infiltrated compressed wood (PICW) (Erickson, 1965; Yano et al., 1997); thermo-mechanical compressed wood (TMCW) (Kutnar et al., 2008); delignified compressed wood (DCW) (Frey et al., 2018; Song et al., 2018; Liu et al., 2019); delignified and partially fiber surface dissolved compressed wood (D&PSDCW) (Khakalo et al., 2019; 2020); delignified resin-infiltrated compressed wood (DRICW) (Chen et al., 2020b). (d) Impact toughness and (e) Shore D hardness of natural wood, mechanically compressed wood, and self-densified wood. All data were results from longitudinal testing of the wood. (f) Compressive stress-strain curves and (g) compressive strength of natural wood, mechanical-compressed wood, and self-densified wood. The testing direction, as shown in the embedded diagram (f), is the longitudinal compression direction of the wood. (h) Longitudinal flexural strength for natural wood, mechanical-compressed wood, and self-densified wood. Longitudinal flexural stress-strain curves for (i) natural wood, (j) mechanical-compressed wood, and (k) self-densified wood. When the bending load axis is parallel to the compression direction, the longitudinal flexural strength is low (~96 MPa), but when parallel to the non-compression direction, it increases significantly (~328 MPa). The direction of the force axis in both wood and self-dense wood has no significant effect on longitudinal flexural strength, showing uniform consistency. PICW, polymer-infiltrated compressed wood; TMCW, thermo-mechanical compressed wood; DCW, delignified compressed wood; D&PSDCW, delignified and partially fiber surface dissolved compressed wood; DRICW, delignified resin-infiltrated compressed wood; NW, natural wood; SW, self-densified wood.

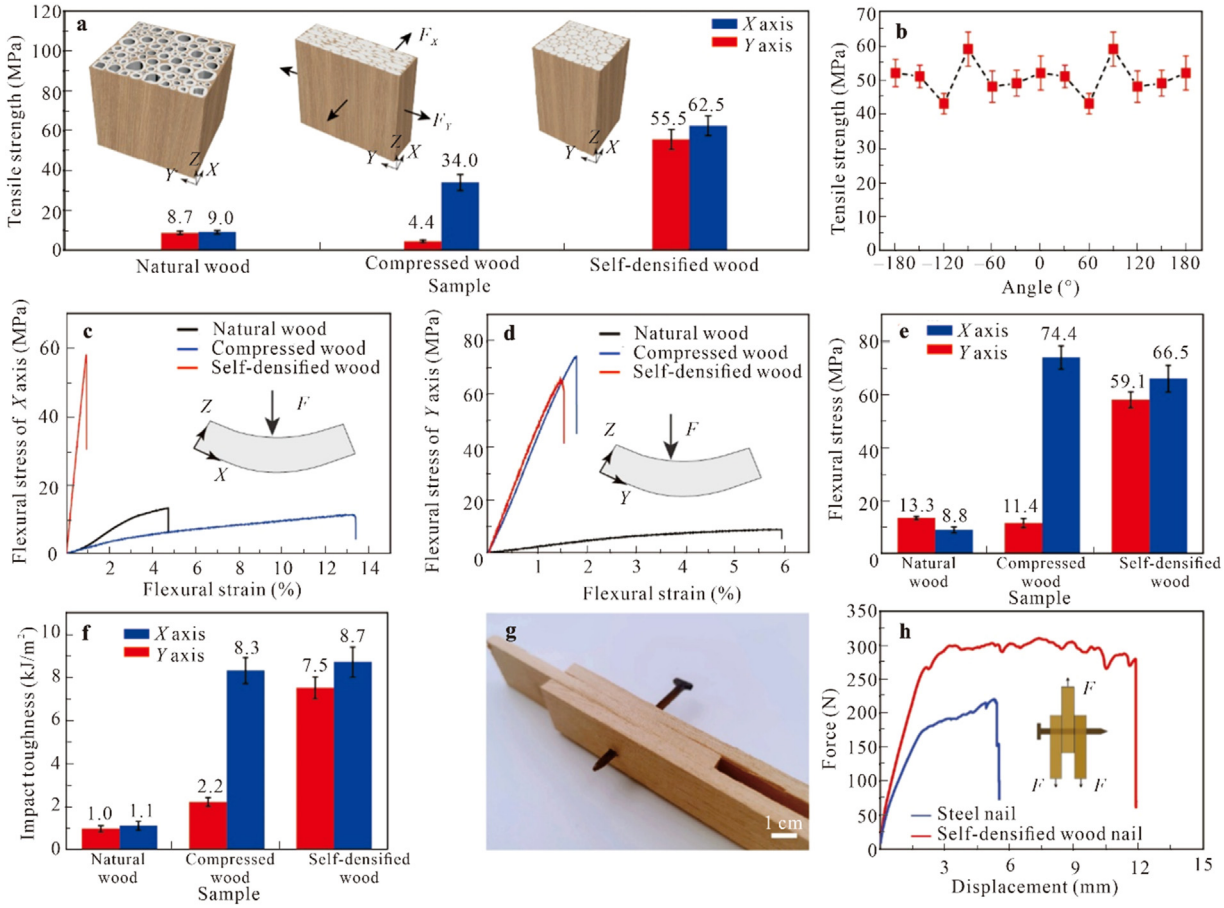

Fig. 4. Transverse section properties of self-densified wood. (a) Schematic diagram and transverse tensile strength of natural wood, mechanically compressed wood, and self-densified wood. (b) Tensile strength of self-densified wood at different angles in the transverse section. Transverse flexural stress-strain curves of natural wood, mechanical-compressed wood, and self-densified wood in (c) *X*-axis and (d) *Y*-axis. (e) Transverse flexural strength and (f) impact toughness of natural wood, mechanical-compressed wood, and self-densified wood in *X*-axis and *Y*-axis directions. (g) Digital images of wood block nailed with the self-densified wood nail. (h) Force-displacement curves of uniaxial tensile test of self-densified wood nail compared to steel nail.

significantly enhances tensile strength but also maintains exceptional toughness, with its fracture work exceeding ten times that of natural wood and its impact toughness of 75.2 kJ/m^2 (Fig. 3d) is six times greater than natural wood, similar to compressed wood. This effectively resolves the conflict between strength and toughness in traditional materials, highlighting its superior mechanical properties, making it highly suitable for high-strength, high-toughness applications (Ritchie, 2011; Huang et al., 2019). The Shore D hardness also significantly increases from 42° to 87° (Fig. 3e). The compressive strength of the self-densified wood is about two times higher than that of natural wood (Figs. 3f and g). The flexural strength of self-dense wood along the growth direction is approximately five times that of natural wood (Fig. 3h). The results of studies have shown that, similar to natural wood, self-dense wood exhibits consistent flexural strength when the bending load is applied parallel to the wood's transverse plane (Hwang et al., 2021). In contrast, compression-dense wood demonstrates significant directional dependence. When the bending load axis is parallel to the compression direction, the longitudinal flexural strength is relatively lower (~96 MPa); however, when the bending load axis is parallel to the non-compression direction, its strength significantly increases to ~328 MPa (Figs. 3h–3k). The remarkable improvement of mechanical properties is attributed to the unique self-densification strengthening at multiple-scales. Self-densified wood maintains a highly densely ordered alignment structure. During the densification process and tensile fracture surface, highly aligned nanocellulose fibers were densely packed and arranged longitudinally (Supplementary Materials, Figs. S7, S13, S18, S23–S27), significantly increasing the interfacial area among nanofibers. At the molecular scale, during the swelling and dehydration processes, the abundant hydroxyl groups in cellulose molecular chains facilitate numerous cycles of hydrogen bond formation, breaking, and reformation. The densification process promotes the formation of numerous hydrogen bonds between the highly aligned cellulose chains. Simultaneously, the microstructure of wood cells transitions from a tubular to a dense columnar configuration, significantly reducing both defect density and cell size. The enhanced hydrogen bond density, combined with the reduction in structural defects and dimensions, synergistically contributes to the ultrahigh strength of self-densified wood. Furthermore, to demonstrate the universality,

balsa wood was also processed by using the same method, resulting in a 14-fold increase in tensile strength (Supplementary Materials, Figs. S28, S29). All these results collectively confirm that the super-strong wood can be obtained through a microstructure regulation strategy without hot-pressing.

3.3. Transverse section reinforcement

We further investigated the tensile behavior in the transverse section to better understand the comprehensive mechanical properties of the self-densified wood. Transverse mechanical compression(perpendicular to the longitudinal direction) is usually essential to fabricate the densified wood. Such unidirectional compression enables the wood cell walls to collapse and become densely packed along the compression direction. This anisotropic microstructure further intensifies the anisotropy of the wood's mechanical properties. The transverse tensile strength perpendicular to the compression direction is significantly improved, while the transverse tensile strength parallel to the compression direction is greatly reduced. Specifically, the tensile strength of compressed wood increases from 9.0 to 34.0 MPa along the *Y*-axis but dramatically decreases from 8.7 to 4.4 MPa along the *X*-axis (Fig. 4a, Supplementary Materials, Figs. S30–S32). In contrast, the self-densified wood is achieved by a uniform shrinkage in the transverse section. This active shrinkage gives the self-densified wood a uniform microstructure. The transverse tensile strength increases by an order of magnitude, reaching 62.5 MPa, nearly seven times that of natural wood, while maintaining high consistency in both the tangential and radial directions across the cross-section (Fig. 4b). Additionally, the flexural strength and impact toughness of self-densified wood also show similar trends. The flexural strength is increased by 7.5 and 7.9 times than that of the original wood in the *X*-axis and *Y*-axis directions, respectively (Figs. 4c–4f). In summary, as shown in Tables S1 and S2, the balanced mechanical enhancement of self-densified wood in all directions significantly mitigates multi-directional failure risks, improving structural safety and durability. This performance breakthrough extends its applications beyond traditional longitudinal load-bearing structures to scenarios requiring transverse and complex mechanical properties. To explore the full potential of the self-densified wood, we fabricated it into a nail for demonstration. As shown in Figs. 4g and 4h, Supplementary Materials, Fig. S33, due to the high strength and sharpness of the densified wooden nails, they can easily penetrate the 3-layer wood without requiring pre-drilling. The load-carrying test shows that it can withstand >300 N and the failure of the sample is caused by the rupture of the wood board, which is even stronger than the steel nail (Figs. 4h and Supplementary Materials, Fig. S34). This result indicates that wood nail has better nail-holding ability and causes less damage to the wood block during nailing process, which is a promising eco-friendly alternative to steel nail for many commercial applications (Ruan et al., 2022).

4. Conclusion

In this work, we demonstrate a self-densified, super-strong wood via the microstructural regulation strategy. Our approach involves releasing cellulose fibrils from the lignin confinement through partial delignification, then the cellulose fibrils moving inward and filling the cell lumen via LiCl/DMAc swelling process, and concluding with air-drying. This unique self-densification process endows the resulting wood with a uniform microstructure, and compared to natural wood, its mechanical properties are significantly and uniformly enhanced in all directions. The improvement in mechanical properties not only compensates for the performance heterogeneity caused by compression in conventional densified wood but also overcomes its limitation of being suitable only for longitudinal load-bearing scenarios. The newly developed self-densified super-strong wood makes it highly attractive for many engineering and structural applications including complex mechanical scenarios. This versatility is demonstrated in our study through the fabrication of a wood nail.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author contributions

Mingwei Zhu, Dafang Huang and Suiyi Li designed the experiments; Jie Li and Dafang Huang contributed to material preparation and most material characterizations, data analysis; Suiyi Li and Dafang Huang contributed to manuscript writing; Jianbing Hu, Zhiru Cao and Yang Guo contributed to material preparation and mechanical tests; and Yu Ding and Yanfeng Chen helped to review manuscript. All authors commented on the final manuscript.

Availability of data

Data will be made available on request.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jobab.2025.03.001.

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